🧠 The Autonomous Outreach Curator: End-to-End Workflow

Version: 14.0 (Autonomous LLM Curation Edition)

The architecture has evolved beyond simple keyword scraping. The Agent now operates as a **Semantic Knowledge Curator**, capable of sourcing, filtering, entirely de-duplicating, and directly extracting elite PR publisher data—without relying on manual keyword inputs.

Here is the exact step-by-step workflow of how the Agent autonomously produces and delivers ultra-high-quality data directly into your Google Sheets.

🏗️ Visual Flowchart: The Data Pipeline

```
flowchart TD
    A[🤖 Initiate Request for High-Quality Sites] --> B[🧠 Phase 1: LLM Knowledge Brainstorming]
    B --> |"Pulls Top-Tier Publishers (MENA, Forbes, Global)"| C[📋 Generates Raw List of 200+ Premium Domains]
    C --> D[🛡️ Phase 2: Global Memory Check]
    D --> |"Checks against master_scraped_domains.txt"| E{Is Domain in Memory?}
    E -->|Yes| F[❌ Silently Destroy Duplicate]
    E -->|No| G[✅ Approve Unique Domain]
    G --> H[⚙️ Phase 3: Format & Categorize]
    H --> |"Forces 'www.' prefix & classifies niche"| I[📧 Phase 4: Dynamic Email Extraction]
    I --> |"Scans /contact, /about, Write-for-Us"| J{Emails Found?}
    J -->|Yes >= 1| K[✂️ Cap strictly at 5 Emails]
    J -->|Zero| L[⚠️ Output: 'No Email Found']
    K --> M[📊 Phase 5: Google Sheets Pipeline]
    L --> M
    M --> N[🚀 Auto-Appends to 'Combined Data' Tab]
    N --> O[🔒 Updates Memory Bank with New Domains]
    O --> P[🏁 Process Complete: 0% Duplication]
```

🔍 Phase 1: Deep Semantic Brainstorming (No Keywords)

Instead of blindly typing narrow keywords into Google (which often returns low-quality spam or agencies), the Agent taps directly into its massive internal LLM (Large Language Model) knowledge base.

1. **Targeting Elite Hubs:** It actively recalls top-tier magazines, premium news networks, and deep-niche business portals across the UAE, Saudi Arabia, Egypt, Qatar, and North Africa.
2. **Quality Guarantee:** By bypassing search engines, it avoids "parasite SEO" blogs and strictly targets sites with high Domain Authority (DA) and organic human traffic.

🛡️ Phase 2: The Immortal Memory Check (0% Duplication)

Before presenting a single website, the Agent actively cross-references the raw brainstormed list against your **Global Memory Bank** (`master_scraped_domains.txt`).

1. **Historical Recall:** This text file contains over **1,250+ domains**, including everything from your original 984-row CSV dataset and all previous Agent drops.
2. **Silent Erasure:** If a newly brainstormed domain exists anywhere in this bank, the Agent natively destroys the result. It only extracts domains that are 100% brand new to you.

⚙️ Phase 3: Data Standardization

The Agent automatically transforms the raw domains into your exact spreadsheet structure.

- **Domain:** Forces standard `www.` prefix (e.g., `www.gqmiddleeast.com`).
- **URL:** Forces full HTTPS formatting.
- **Category:** Assigns a highly specific niche identifier (e.g., "Men's Lifestyle & Business", "Oman Economic Review", "Tech & Startup News").

📧 Phase 4: Intelligent Native Email Extraction

The Agent launches a Python sub-routine to aggressively scan the newly discovered domains for contact details.

1. **Prioritized Crawling:** It visits `/contact`, `/team`, `/about`, and any `Write for Us` pages natively.
2. **Spam Filtering:** It parses `mailto:` links and plaintext using Regex, strictly keeping emails that match the root domain (or generic hosts like `@gmail.com`) while erasing 3rd-party software emails (`@sentry.io`).
3. **Hard Capping:** It stops immediately upon compiling a maximum of **5 valid emails per domain** to prevent messy data outputs.

🚀 Phase 5: Google Sheets API Auto-Deployment

Without requiring a single manual copy/paste from the user, the Agent:

1. Authenticates securely via the Google Cloud Service Account (`.json` credential).
2. Connects live to your `Combined Data` Google Sheet.
3. Automatically appends the freshly curated, de-duplicated, and email-enriched columns directly onto the bottom of your sheet.
4. **Final Lock:** Submits every newly deployed domain permanently into the Global Memory Bank so it can never be used again.

> ***Result:*** *A continuous, frictionless loop generating elite, highly verified publisher outreach data with absolutely zero crossover or manual intervention.*